Sixth Allocation:

	C	H	K	P	a_i
T	6	(5) 8	8	(25) 5	30/5/0
B	(35) 5	(5) 11	9	7	40/5/0
M	8	(18) 9	(32) 7	13	50/18/0
b_j	35/0	28/23/5/0	32/0	25/0	

Transportation schedule :

$T \rightarrow H$, $T \rightarrow P$, $B \rightarrow C$, $B \rightarrow H$, $M \rightarrow H$, $M \rightarrow K$

The total Transportation cost = (5×8) + (25×5)+ (35×5) + (5×11)+ (18×9) + (32×7)

= 40+125+175+55+162+224

= ₹ 781

Method: 3 Vogel's Approximation Method (VAM)

Vogel's approximation method yields an initial basic feasible solution which is very close to the optimum solution.Various steps involved in this method are summarized as under

Step 1: Calculate the penalties for each row and each column. Here penalty means the difference between the least and the next higher cost in a row and in a column.

Step 2: Select the row or column with the largest penalty.

Step 3: In the selected row or column, allocate the maximum feasible quantity to the cell with the minimum cost.

Step 4: Eliminate the row or column where all the allocations are made.

Step 5: Write the reduced transportation table and repeat the steps 1 to 4.

Step 6: Repeat the procedure until all the allocations are made.

DO YOU KNOW?

Optimum solution for any transportation problem can be obtained by using MODI method.

Example 10.5

Find the initial basic feasible solution for the following transportation problem by VAM

		Distribution Centers				Availability
		D_1	D_2	D_3	D_4	
origin	S_1	11	13	17	14	250
	S_2	16	18	14	10	300
	S_3	21	24	13	10	400
Require-ment		200	225	275	250	

Solution:

Here $\sum a_i = \sum b_j = 950$

(i.e) Total Availability = Total Requirement

∴ The given problem is balanced transportation problem.

Hence there exists a feasible solution to the given problem.

First let us find the difference (penalty) between the first two smallest costs in each row and column and write them in brackets against the respective rows and columns.

	D_1	D_2	D_3	D_4	a_i	Penalty
S_1	(200) 11	13	17	14	250/50	2
S_2	16	18	14	10	300	4
S_3	21	24	13	10	400	3
b_j	200/0	225	275	250		
Penalty	(5)	5	1	4		

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